

I can't swallow

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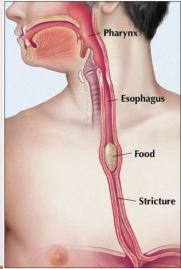
Learning Objectives

- Definition & Classification
- Anatomy & Physiology
- Approach to dysphagia
- Differential diagnosis
- Clinical history
- Specific investigation
- Case scenarios



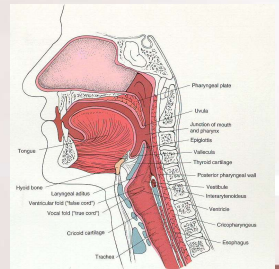
Definitions & Classifications

- **Dysphagia** - Greek terms *dys* meaning 'disordered' or 'ill', and *phago* meaning 'eat' or 'swallow'
 - The function of **clearing food and drink** through the oral cavity, pharynx and oesophagus into the stomach at an **appropriate rate and speed**
- **Oropharyngeal dysphagia**
 - Difficulty with the **initial phases** of a swallow, from mouth to esophagus
- **Esophageal dysphagia**
 - **Sensation** that foods and or liquids are being obstructed in their passage from the mouth to the stomach



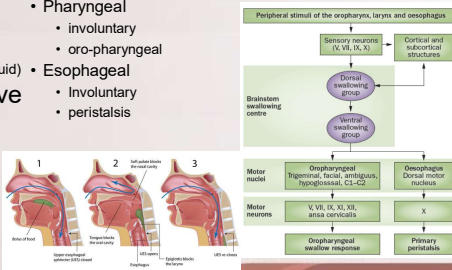
Anatomy

- Tongue, oropharynx, Upper esophageal sphincter, upper 5% of esophagus: **striated muscle**
- Distal 50-60% of esophagus: **smooth muscle**
- Mid 35%-40% of esophagus: **mixed**



Physiology: Phase of swallowing

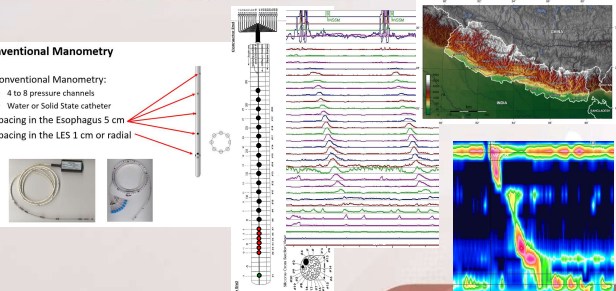
- Oral preparatory
 - Voluntary
 - Mastication (Solid)
 - Glossopalatal seal (fluid)
- Oral propulsive
 - Mouth → Pharynx → Esophagus
 - Tongue movement
- Pharyngeal
 - involuntary
 - oro-pharyngeal
- Esophageal
 - Involuntary
 - peristalsis

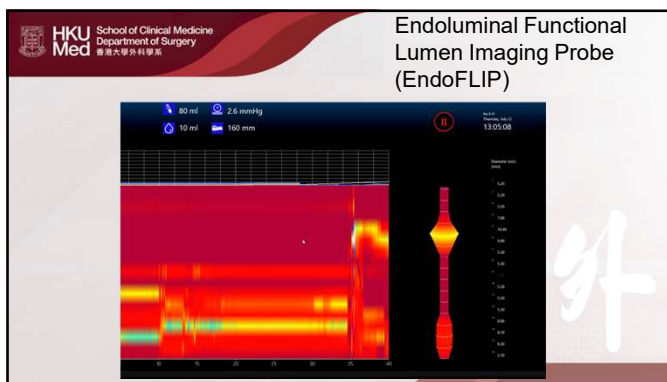
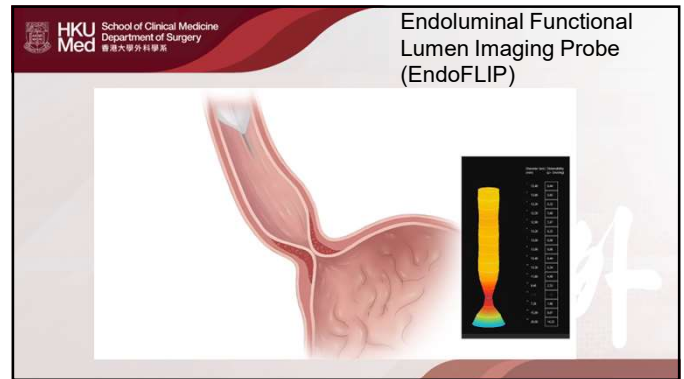
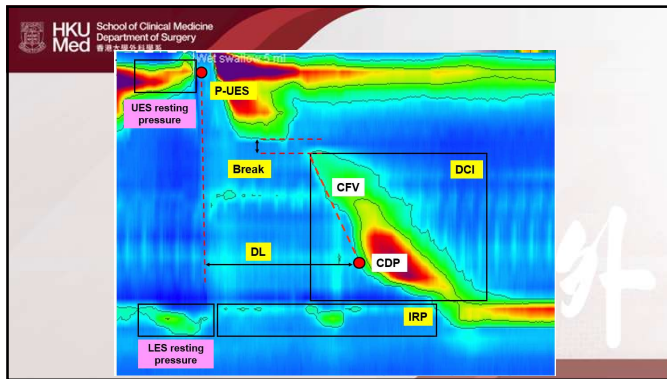


Manometry to HRM

Conventional Manometry

- Conventional Manometry:
 - 4 to 8 pressure channels
 - Water or Solid State catheter
 - Spacing in the Esophagus 5 cm
 - Spacing in the LES 1 cm or radial





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Approach to dysphagia

- Real dysphagia?
 - Globus hystericus (something sticking in the throat)
- Oro-pharyngeal vs. esophageal dysphagia
- Mechanical (anatomic) vs. functional (motility)

****Esophageal dysphagia patient may perceive the location at cervical region mimicking oropharyngeal dysphagia**

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Oropharyngeal Dysphagia

- Difficulty initiating a swallow, repetitive swallowing
- Nasal regurgitation
- Coughing, diminished cough reflex
- Drooling of saliva or food
- Choking
- Dysarthria & diplopia
- Halitosis
- Recurrent pneumonia

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Esophageal Dysphagia

TABLE 2. Most Common Causes of Esophageal Dysphagia

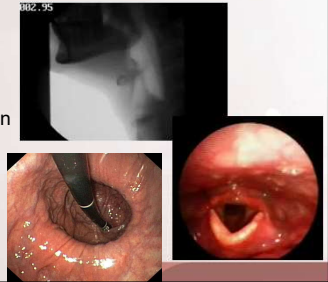
Types	Conditions
Intraluminal causes	Foreign bodies (acute dysphagia)
Mediastinal diseases—obstruct the esophagus by direct invasion, compression, or through lymph-node enlargement	Tumors (eg, lung cancer, lymphoma)
	Infections (eg, tuberculosis, histoplasmosis)
	Cardiovascular (dilated aortices, vascular compression)
Mucosal diseases—narrow the lumen through inflammation, fibrosis, or neoplasia	Peptic stricture secondary to gastroesophageal reflux disease
	Esophageal rings and webs (scleroderma dysphagia or Plummer-Vinson syndrome) ¹
	Esophageal tumors
	Chemical injury (eg, caustic ingestion, pill esophagitis, sclerotherapy for varices)
	Radiation injury
	Infectious esophagitis (eg, herpes virus, <i>Candida albicans</i>)
	Eosinophilic esophagitis
	Tumor or granulation overgrowth in esophageal stenosis
Neuromuscular diseases—alter the esophageal smooth muscle and its innervation, disrupting peristalsis or lower esophageal sphincter relaxation, or both	Achalasia (idiopathic and associated with neoplasia, Chagas disease, other)
	Scleroderma, mixed connective tissue diseases (myositis)
	Esophageal spasms (jackhammer esophagus)
Postsurgical	After fundoplication, antireflux devices

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Esophageal Dysphagia		
History	Mechanical	Function
Onset	Gradual or Sudden	Usually gradual
Progression	Often	Variable
Type of bolus	Solid	Solid or liquids
Response to bolus	Often regurgitation	Usually passes with drinking liquid or repeated swallowing
Temperature	No	May vary with temperature of food

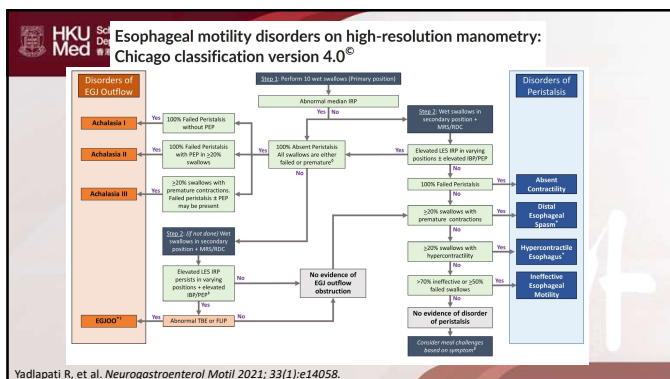
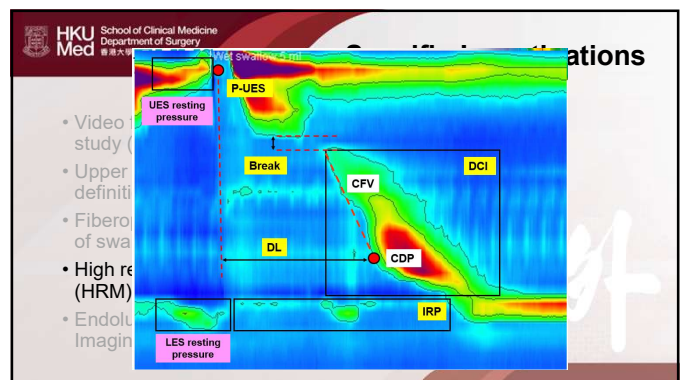
Specific Investigations

- Video fluoroscopy swallowing study (VFSS), Barium swallow
- Upper endoscopy (High definition)
- Fiberoptic endoscopic evaluation of swallowing (FEES)
- High resolution manometry (HRM)
- Endoluminal Functional Lumen Imaging Probe (EndoFLIP)



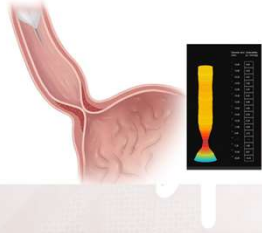
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Scenario

- 35/M. Non smoker & non drinker. Chest pain, regurgitation, progressive on & off dysphagia for 2 year. Required flushing with fluid. Weight loss of 20lbs.
- Achalasia
- Reflux Stricture
- Stroke
- Ca esophagus
- Distal esophageal spasm

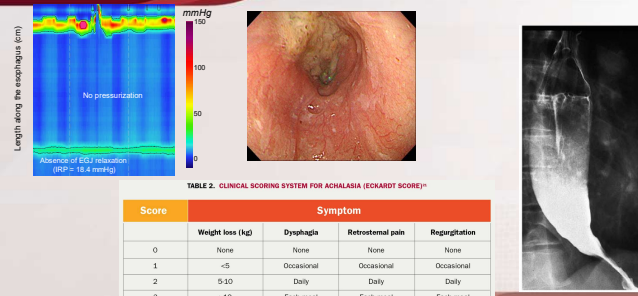
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Scenario

- 75/M, Chiu Chow Ancestry, Chronic Smoker & Drinker. Progressive dysphagia for 1 month. Weight loss. Regurgitating solid food. PE: Cachexic, R. SCF LN.
- Achalasia
- Reflux Stricture
- Stroke
- Ca esophagus
- Distal esophageal spasm

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Achalasia

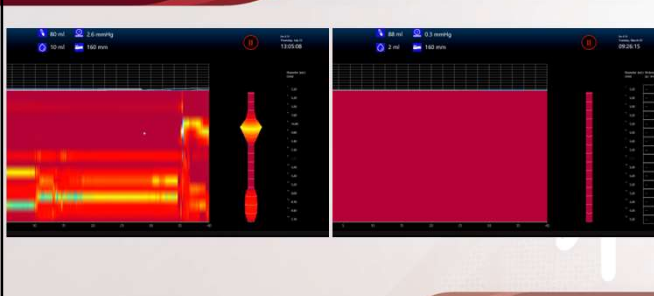


Length along the esophagus (cm) mmHg
No pressurization
Absence of LES relaxation (IRP = 16.4 mmHg)

TABLE 2. CLINICAL SCORING SYSTEM FOR ACHALASIA (ECKART SCORE)¹⁰

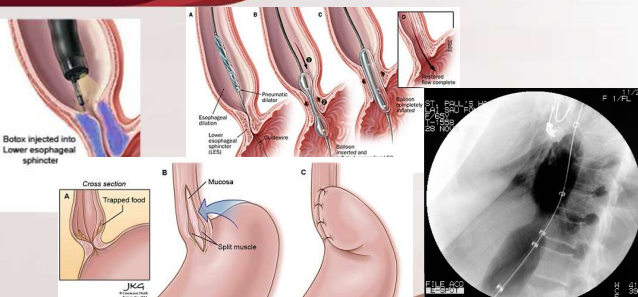
Score	Symptom			
	Weight loss (kg)	Dysphagia	Retrosternal pain	Regurgitation
0	None	None	None	None
1	<5	Occasional	Occasional	Occasional
2	5-10	Daily	Daily	Daily
3	>10	Each meal	Each meal	Each meal

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Treatment



Bolus injected into Lower esophageal sphincter

Cross section

A Trapped food

B Mucosa

C Split muscle

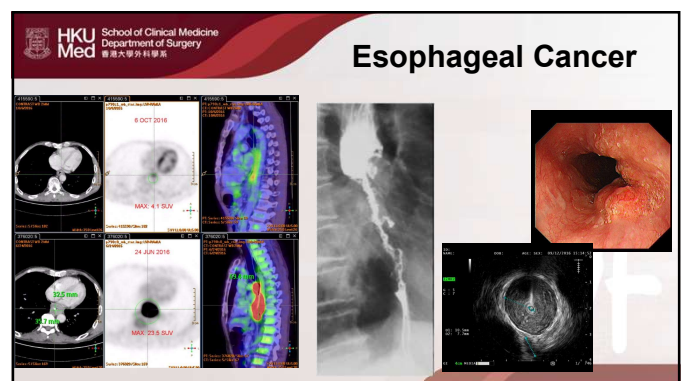
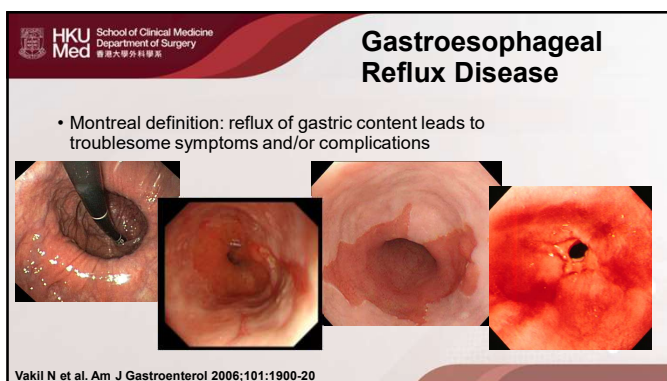
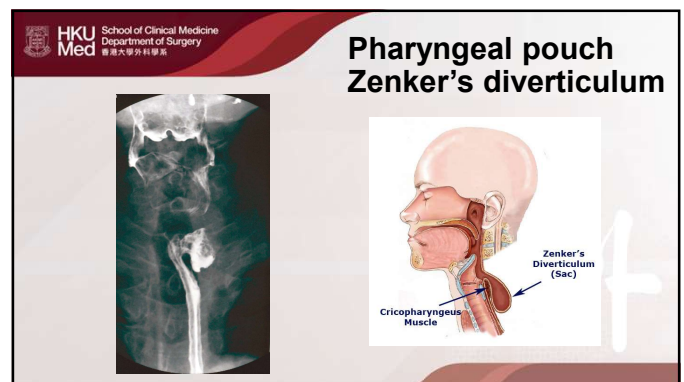
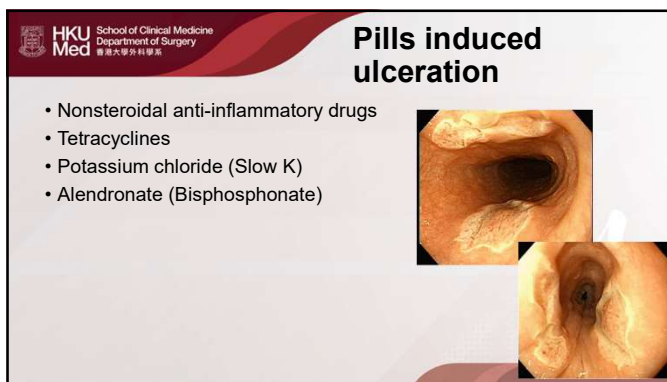
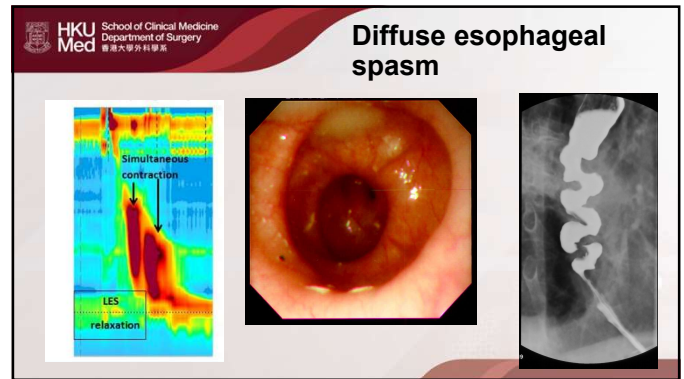
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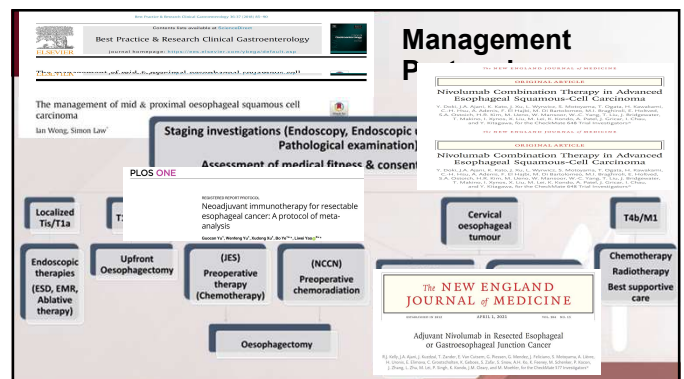
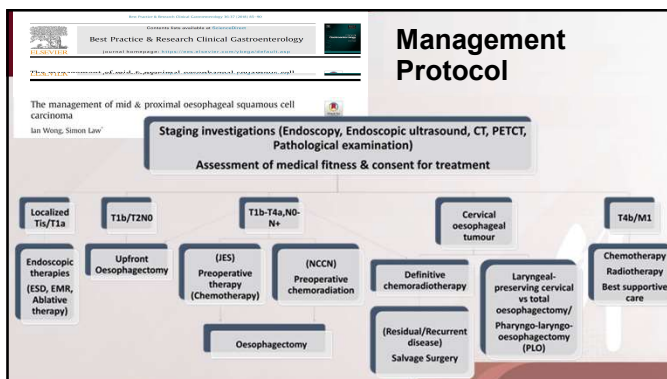
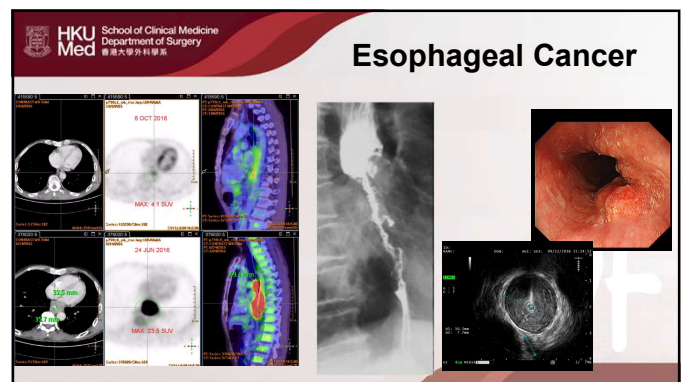
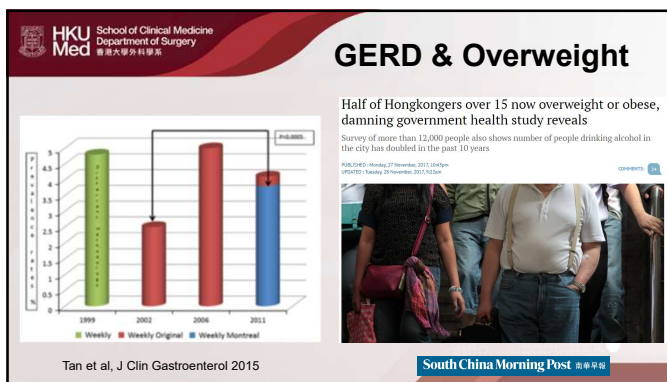
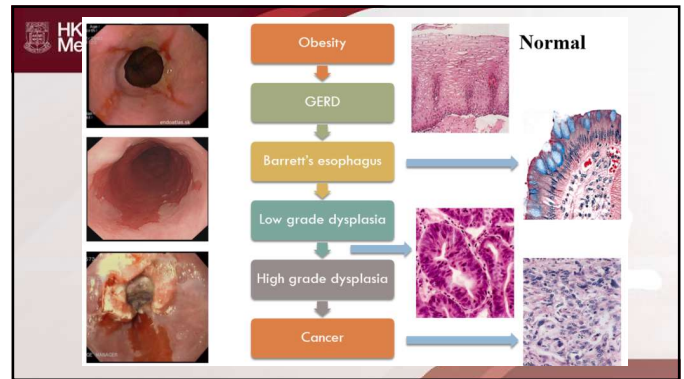
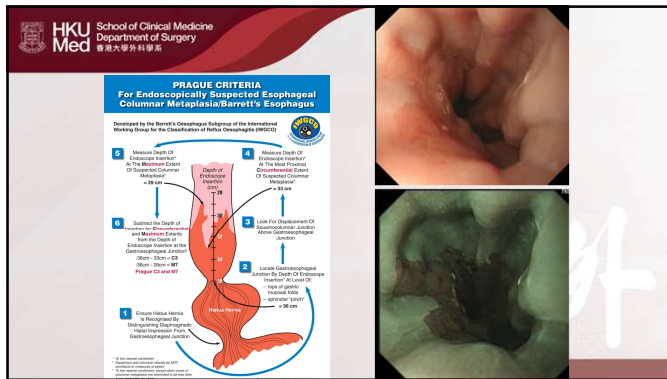
Peroral Endoscopic Myotomy (POEM)



① ② ③ ④ ⑤

Inoue H et al. POEM for esophageal achalasia. Endoscopy 2010; 42: 265-271.





Operation Esophagectomy

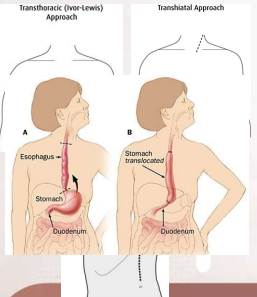
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- Phase: II (Lewis-Tanner), III (McKeown)
- Field of lymph node dissection:
 - II (Thorax + Abdomen), III (II + Neck)
- Conduit: Stomach, Colon, Jejunum
- Route: Orthotopic, Retrosternal, Subcutaneous
- Anastomosis: Stapled (Linear, Circular), Handsewn
- Approach: Open, VATS, Total MIE, Robotic
- Position: Supine, Left lateral, Prone



Torek, Surg Gynecol Obstet, 1913

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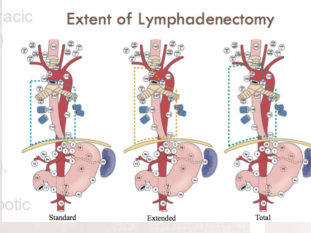
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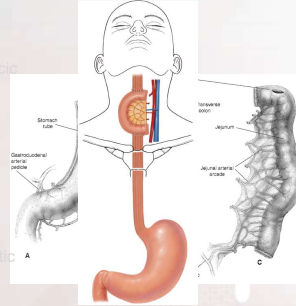
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Extent of Lymphadenectomy



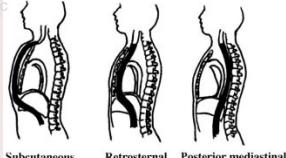
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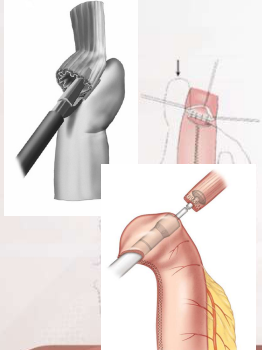
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Subcutaneous Retrosternal Posterior mediastinal

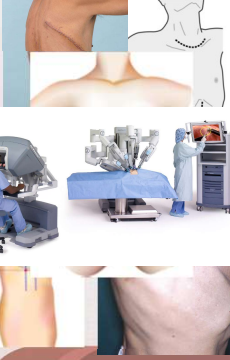
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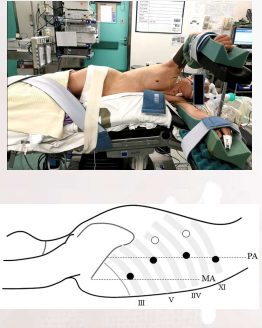
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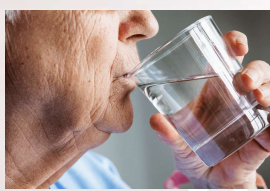
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TELL US WHAT
MBBS Whole Class Session
YOU THINK

THANK YOU!



Look for today's teacher **IYH WONG**
and the others you may have missed.
(HKU Portal login required)